

## 6.8 Special Determinants

A determinant of size  $n$  has  $n!$  terms, so a general formula for a large matrix is hopeless. Some matrices, though, have so much structure that their determinants collapse to a short product or a recurrence. This section computes the most useful of them: the Vandermonde determinant, which governs polynomial interpolation; tridiagonal determinants, which satisfy a three-term recurrence; block triangular determinants, which split into smaller ones; and the determinant of a rank-one update of the identity. The proofs share one idea: find the structure first (a polynomial with known roots, a row with two non-zero entries, a pattern of zeros) and let the defining properties of the determinant do the rest.

### 6.8.1 The Vandermonde determinant

Chapter 2 proved that there is exactly one polynomial of degree at most  $n$  through  $n + 1$  points with distinct first coordinates (Theorem 2.7.3). Behind it sat a square matrix of powers of the nodes. Its determinant turns out to be a product of all the differences of the nodes, and that product explains at a glance why distinct nodes are exactly what is needed.

For  $x_1, \dots, x_n \in F$ , the **Vandermonde matrix** is

$$\mathbf{V}(x_1, \dots, x_n) := \begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{pmatrix} \in M_n(F),$$

whose  $(i, j)$ -entry is  $x_i^{j-1}$ . Row  $i$  lists the powers  $1, x_i, \dots, x_i^{n-1}$  of the  $i$ -th node. For  $n = 1$  it is the  $1 \times 1$  matrix  $(1)$ , and for  $n = 2$  its determinant is  $x_2 - x_1$ .

#### Theorem 6.8.1 — Vandermonde Determinant

Let  $n \geq 1$  and  $x_1, \dots, x_n \in F$ . Then

$$\det \mathbf{V}(x_1, \dots, x_n) = \prod_{1 \leq i < j \leq n} (x_j - x_i),$$

where the empty product (for  $n = 1$ ) is 1.

*Idea.* Induction on  $n$ . Replace the last node  $x_n$  by a variable  $t$ . Expanding along the last row shows the determinant is a polynomial  $f(t)$  of degree at most  $n - 1$ , with leading coefficient the smaller Vandermonde determinant. Setting  $t = x_i$  for  $i < n$  makes two rows equal, so  $x_1, \dots, x_{n-1}$  are roots of  $f$ . A polynomial of degree at most  $n - 1$  with these  $n - 1$  roots is its leading coefficient times  $\prod_{i < n} (t - x_i)$ . Now put  $t = x_n$  back. The roots argument needs the  $x_i$  distinct; if they are not, both sides are 0 anyway.

*Proof.* We use induction on  $n$  (Theorem 0.2.7). For  $n = 1$ ,  $\det(1) = 1$ , the empty product.

Let  $n \geq 2$ , assume the formula for  $n - 1$  nodes, and write  $\mathbf{V} = \mathbf{V}(x_1, \dots, x_n)$ . Let  $C_{n1}, \dots, C_{nn}$  be the cofactors of  $\mathbf{V}$  along row  $n$  (Definition 6.6.1). Each  $C_{nj}$  is computed

after deleting row  $n$ , so it does not involve  $x_n$ . Define

$$f(t) := \sum_{j=1}^n C_{nj} t^{j-1} \in F[t].$$

For  $c \in F$ , let  $\mathbf{V}_c$  be  $\mathbf{V}$  with row  $n$  replaced by  $(1, c, \dots, c^{n-1})$ . Then  $\mathbf{V}_c$  has the same cofactors along row  $n$  as  $\mathbf{V}$ , so by Theorem 6.6.1 along row  $n$ ,

$$\det \mathbf{V}_c = \sum_{j=1}^n c^{j-1} C_{nj} = f(c). \quad (*)$$

In particular  $\det \mathbf{V} = f(x_n)$ . The coefficient of  $t^{n-1}$  in  $f$  is  $C_{nn} = (-1)^{2n} \det \mathbf{V}(x_1, \dots, x_{n-1})$ , since deleting row  $n$  and column  $n$  of  $\mathbf{V}$  leaves the Vandermonde matrix of the first  $n-1$  nodes.

*Case 1: two of  $x_1, \dots, x_{n-1}$  are equal, say  $x_p = x_q$  with  $p < q < n$ .* Then rows  $p$  and  $q$  of  $\mathbf{V}$  are equal. By Theorem 6.4.5,  $\det \mathbf{V} = \det \mathbf{V}^\top$ , and  $\mathbf{V}^\top$  has two equal columns, so  $\det \mathbf{V} = 0$  because the determinant is alternating (Definition 6.2.2). The product also vanishes, since it contains the factor  $x_q - x_p = 0$ .

*Case 2:  $x_1, \dots, x_{n-1}$  are distinct.* For  $i < n$ , the matrix  $\mathbf{V}_{x_i}$  has rows  $i$  and  $n$  equal, so  $f(x_i) = \det \mathbf{V}_{x_i} = 0$  by  $(*)$  and the argument of Case 1. Hence  $x_1, \dots, x_{n-1}$  are  $n-1$  distinct roots of  $f$ , and by Corollary 5.1.4,

$$g(t) := (t - x_1)(t - x_2) \cdots (t - x_{n-1}) \text{ divides } f.$$

Write  $f = hg$  with  $h \in F[t]$ . If  $h \neq 0$ , then  $\deg h = \deg f - (n-1) \leq 0$  by Theorem 0.8.3, since  $\deg f \leq n-1$ ; so in every case  $h$  is a constant  $h \in F$ . As  $g$  is monic of degree  $n-1$ , the coefficient of  $t^{n-1}$  in  $hg$  is  $h$ , so  $h = C_{nn} = \det \mathbf{V}(x_1, \dots, x_{n-1})$ . Evaluating at  $x_n$  (Theorem 0.8.5) and using the induction hypothesis,

$$\begin{aligned} \det \mathbf{V} &= f(x_n) = \det \mathbf{V}(x_1, \dots, x_{n-1}) \prod_{i=1}^{n-1} (x_n - x_i) \\ &= \prod_{1 \leq i < j \leq n-1} (x_j - x_i) \prod_{i=1}^{n-1} (x_n - x_i). \end{aligned}$$

The two products together run over all pairs  $i < j \leq n$ , the second one collecting exactly the pairs with  $j = n$ . This proves the formula for  $n$ , and the theorem follows by induction. ■

### Example. (A $3 \times 3$ Vandermonde Determinant)

Compute  $\det \mathbf{V}(1, 2, 4)$  from the formula and by cofactor expansion.

*Solution.* The formula gives  $(2-1)(4-1)(4-2) = 1 \cdot 3 \cdot 2 = 6$ . Directly, expanding

$$\mathbf{V}(1, 2, 4) = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 4 & 16 \end{pmatrix}$$

along the first row gives  $1 \cdot (32 - 16) - 1 \cdot (16 - 4) + 1 \cdot (4 - 2) = 16 - 12 + 2 = 6$ .

The product vanishes exactly when a factor does, because a field has no zero divisors. So:

### Corollary 6.8.2 — Distinct Nodes Give an Invertible Vandermonde Matrix

$\mathbf{V}(x_1, \dots, x_n)$  is invertible if and only if  $x_1, \dots, x_n$  are **distinct**.

*Proof.* In a field, a product is non-zero if and only if every factor is non-zero (Theorem 0.7.1 (f), applied repeatedly). So by Theorem 6.8.1,  $\det \mathbf{V}(x_1, \dots, x_n) \neq 0$  if and only if  $x_j \neq x_i$  for all  $i < j$ . By Theorem 6.5.2, this is equivalent to invertibility. ■

Now the promised two-line proof of Theorem 2.7.3. Given distinct  $x_0, \dots, x_n$  and values  $y_0, \dots, y_n$ , a polynomial  $p = c_0 + c_1x + \dots + c_nx^n$  satisfies  $p(x_i) = y_i$  for all  $i$  if and only if  $\mathbf{V}(x_0, \dots, x_n) \mathbf{c} = \mathbf{y}$ , since row  $i$  of this system reads  $c_0 + c_1x_i + \dots + c_nx_i^n = y_i$ . By Corollary 6.8.2 the matrix is invertible, so the system has exactly one solution  $\mathbf{c} = \mathbf{V}(x_0, \dots, x_n)^{-1} \mathbf{y}$  (Theorem 2.4.4), and the interpolating polynomial exists and is unique.

*Remark.* This is not a proof from nothing. The root bound of Chapter 5 did the real work inside Theorem 6.8.1, exactly as the root bound did in Chapter 2. What the determinant adds is an explicit number: it says how close to singular the interpolation problem is, since nodes that nearly coincide make the product small.

*Warning. Watch the shape and the order.* The formula is for rows  $(1, x_i, \dots, x_i^{n-1})$  starting at the power 0, and for the factors  $x_j - x_i$  with  $i < j$ , “later node minus earlier node”. Dropping the column of ones changes the answer:  $\det \begin{pmatrix} x_1 & x_1^2 \\ x_2 & x_2^2 \end{pmatrix} = x_1x_2(x_2 - x_1)$ , not  $x_2 - x_1$ . Writing the factors as  $x_i - x_j$  instead flips the sign of each of the  $\binom{n}{2}$  factors, so it multiplies the answer by  $(-1)^{\binom{n}{2}}$ : already for  $n = 2$  and  $n = 3$  this is  $-1$ , a wrong sign.

*Quick check.* Without expanding, find  $\det \mathbf{V}(1, 2, 3, 5)$ .

*Solution.* The pairs  $i < j$  give  $(2-1)(3-1)(5-1)(3-2)(5-2)(5-3) = 1 \cdot 2 \cdot 4 \cdot 1 \cdot 3 \cdot 2 = 48$ .

## 6.8.2 Tridiagonal determinants

Many matrices in applications have non-zero entries only on the diagonal and next to it: discretized differential equations, chains of springs, the recurrences of orthogonal polynomials.

Such a matrix is **tridiagonal**. Its determinant is not a closed formula in general, but it obeys a recurrence in the size, because the last row has only two non-zero entries.

Let  $\mathbf{T}_n \in M_n(F)$  have diagonal entries  $a_1, \dots, a_n$ , entries  $b_1, \dots, b_{n-1}$  just above the diagonal ( $(i, i+1)$ -entry  $b_i$ ), entries  $c_1, \dots, c_{n-1}$  just below it ( $(i+1, i)$ -entry  $c_i$ ), and zeros elsewhere:

$$\mathbf{T}_n = \begin{pmatrix} a_1 & b_1 & & & \\ c_1 & a_2 & b_2 & & \\ & c_2 & \ddots & \ddots & \\ & & \ddots & a_{n-1} & b_{n-1} \\ & & & c_{n-1} & a_n \end{pmatrix}.$$

For  $1 \leq k \leq n$ , the top-left  $k \times k$  block of  $\mathbf{T}_n$  is the tridiagonal matrix  $\mathbf{T}_k$  built from  $a_1, \dots, a_k$ ,  $b_1, \dots, b_{k-1}$ ,  $c_1, \dots, c_{k-1}$ . Write  $D_k = \det \mathbf{T}_k$ , and set  $D_0 = 1$ .

### Theorem 6.8.3 — Tridiagonal Recurrence

With this notation,  $D_1 = a_1$ , and for  $2 \leq k \leq n$ ,

$$D_k = a_k D_{k-1} - b_{k-1} c_{k-1} D_{k-2}.$$

*Proof.*  $D_1 = \det(a_1) = a_1$ . For  $k = 2$ ,  $D_2 = a_1 a_2 - b_1 c_1 = a_2 D_1 - b_1 c_1 D_0$ . Let  $k \geq 3$ . Row  $k$  of  $\mathbf{T}_k$  has only two possibly non-zero entries,  $c_{k-1}$  in column  $k-1$  and  $a_k$  in column  $k$ . By Theorem 6.6.1 along row  $k$ ,

$$D_k = c_{k-1}(-1)^{2k-1} M_{k,k-1} + a_k(-1)^{2k} M_{kk} = a_k M_{kk} - c_{k-1} M_{k,k-1},$$

where  $M_{kk} = \det \mathbf{T}_{k-1} = D_{k-1}$ . The minor  $M_{k,k-1}$  is the determinant of the  $(k-1) \times (k-1)$  matrix obtained by deleting row  $k$  and column  $k-1$ . Its last column consists of rows  $1, \dots, k-1$  of column  $k$  of  $\mathbf{T}_k$ , which is  $b_{k-1}$  in the last position and 0 elsewhere. Expanding along this last column (Theorem 6.6.1), with sign  $(-1)^{(k-1)+(k-1)} = 1$ , gives  $M_{k,k-1} = b_{k-1} \det \mathbf{T}_{k-2} = b_{k-1} D_{k-2}$ , because deleting the last row and column leaves rows and columns  $1, \dots, k-2$  of  $\mathbf{T}_k$ . Substituting proves the recurrence. ■

### Example. (Fibonacci Numbers as Determinants)

Let  $\mathbf{T}_n$  have 1 on the diagonal, 1 above it and  $-1$  below it. Find  $D_n = \det \mathbf{T}_n$ .

*Solution.* Here  $a_k = 1$  and  $b_{k-1} c_{k-1} = 1 \cdot (-1) = -1$ , so Theorem 6.8.3 gives  $D_k = D_{k-1} + D_{k-2}$  for  $k \geq 2$ , with  $D_0 = 1$  and  $D_1 = 1$ . Hence

$$D_1, D_2, D_3, \dots = 1, 2, 3, 5, 8, 13, 21, 34, \dots,$$

the Fibonacci numbers:  $D_n = F_{n+1}$ , where  $F_1 = F_2 = 1$  and  $F_k = F_{k-1} + F_{k-2}$ . For instance

$$D_3 = \det \begin{pmatrix} 1 & 1 & 0 \\ -1 & 1 & 1 \\ 0 & -1 & 1 \end{pmatrix} = 1 \cdot (1 + 1) - 1 \cdot (-1 - 0) + 0 = 3.$$

Only the products  $b_{k-1}c_{k-1}$  enter the recurrence, never  $b_{k-1}$  and  $c_{k-1}$  separately. For example, for a real number  $t$ , putting  $2t$  on the diagonal and 1 on both neighboring lines gives  $D_k = 2t D_{k-1} - D_{k-2}$ , so  $D_1 = 2t$ ,  $D_2 = 4t^2 - 1$ ,  $D_3 = 8t^3 - 4t$ . These are the Chebyshev polynomials of the second kind.

### 6.8.3 Block triangular matrices

A matrix can be cut into blocks by a horizontal and a vertical line. When the block in the lower left is zero, the determinant should see only the two square blocks on the diagonal, just as a triangular determinant sees only diagonal entries. This result is used constantly, for instance in Chapter 7 for Schur complements.

#### Theorem 6.8.4 — Block Triangular Determinant

Let  $\mathbf{A} \in M_k(F)$ ,  $\mathbf{D} \in M_l(F)$  and  $\mathbf{B} \in M_{k \times l}(F)$ , with  $k, l \geq 1$ . Then

$$\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ 0 & \mathbf{D} \end{pmatrix} = \det \mathbf{A} \cdot \det \mathbf{D},$$

where 0 is the  $l \times k$  zero matrix. The same holds for  $\begin{pmatrix} \mathbf{A} & 0 \\ \mathbf{C} & \mathbf{D} \end{pmatrix}$  with  $\mathbf{C} \in M_{l \times k}(F)$ .

*Idea.* In the Leibniz formula for the whole matrix, a term picks one entry from each column. The first  $k$  columns have zeros below row  $k$ , so a non-zero term must pick rows  $1, \dots, k$  for the first  $k$  columns, and is then forced to pick rows  $k+1, \dots, n$  for the last  $l$  columns. Such a permutation is a permutation of each block, its inversions are the inversions inside the blocks, and the Leibniz sum factors into two Leibniz sums.

*Proof.* Let  $n = k + l$  and  $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ 0 & \mathbf{D} \end{pmatrix} = (m_{ij})$ . By Definition 6.4.1,  $\det \mathbf{M} = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{j=1}^n m_{\sigma(j)j}$ . Since  $m_{ij} = 0$  for  $i > k \geq j$ , the term of  $\sigma$  is 0 unless  $\sigma(j) \leq k$  for every  $j \leq k$ . For such  $\sigma$ , the injective map  $\sigma$  sends  $\{1, \dots, k\}$  into itself, hence onto itself (a finite set), and so it sends  $\{k+1, \dots, n\}$  onto itself. Therefore  $\sigma$  determines  $\alpha \in S_k$  and  $\beta \in S_l$  by

$$\alpha(j) = \sigma(j) \quad (j \leq k), \quad \beta(q) = \sigma(k+q) - k \quad (q \leq l),$$

and every pair  $(\alpha, \beta)$  arises from exactly one such  $\sigma$ .

We compare signs through inversions (Definition 6.3.1). A pair  $i < j$  with  $i, j \leq k$  is an inversion of  $\sigma$  exactly when it is one of  $\alpha$ ; a pair  $k+p < k+q$  is an inversion of  $\sigma$  exactly when  $p < q$  is one of  $\beta$ ; and a pair  $i \leq k < j$  is never an inversion, since  $\sigma(i) \leq k < \sigma(j)$ . By Definition 6.3.2,  $\text{sgn}(\sigma) = \text{sgn}(\alpha) \text{sgn}(\beta)$ . Also  $m_{\sigma(j)j} = a_{\alpha(j)j}$  for  $j \leq k$  and  $m_{\sigma(k+q), k+q} = d_{\beta(q)q}$ . Hence

$$\det \mathbf{M} = \sum_{\alpha \in S_k} \sum_{\beta \in S_l} \text{sgn}(\alpha) \text{sgn}(\beta) \prod_{j=1}^k a_{\alpha(j)j} \prod_{q=1}^l d_{\beta(q)q} = \det \mathbf{A} \cdot \det \mathbf{D},$$

using Definition 6.4.1 twice. For the lower block triangular case, the transpose of

$\begin{pmatrix} \mathbf{A} & 0 \\ \mathbf{C} & \mathbf{D} \end{pmatrix}$  is  $\begin{pmatrix} \mathbf{A}^\top & \mathbf{C}^\top \\ 0 & \mathbf{D}^\top \end{pmatrix}$ , so by Theorem 6.4.5 and the first case its determinant is  $\det \mathbf{A}^\top \det \mathbf{D}^\top = \det \mathbf{A} \det \mathbf{D}$ .

■

The proof uses only sums and products of entries, so, like the results flagged in §4–§6, it holds for matrices over any commutative ring; §10 uses this over  $F[x]$ . By induction on the number of blocks, a matrix with square blocks  $\mathbf{A}_1, \dots, \mathbf{A}_r$  on the diagonal and zeros below them has determinant  $\det \mathbf{A}_1 \cdots \det \mathbf{A}_r$ ; Theorem 6.4.6 is the case where every block is  $1 \times 1$ .

For example,

$$\det \begin{pmatrix} 2 & 1 & 5 & -3 \\ 1 & 1 & 0 & 7 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 4 & 2 \end{pmatrix} = \det \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \det \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix} = 1 \cdot 2 = 2,$$

and the entries 5, −3, 0, 7 of the upper right block play no role.

**Warning.** There is no “block  $ad-bc$ ” rule. For a general block matrix,  $\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} \neq \det \mathbf{A} \det \mathbf{D} - \det \mathbf{B} \det \mathbf{C}$ . Take  $\mathbf{A} = \mathbf{D} = 0$  and  $\mathbf{B} = \mathbf{C} = \mathbf{I}_2$ : the  $4 \times 4$  matrix  $\begin{pmatrix} 0 & \mathbf{I}_2 \\ \mathbf{I}_2 & 0 \end{pmatrix}$  is the permutation matrix of  $(1\ 3)(2\ 4)$ , a product of two transpositions, so its determinant is +1; the rule would give  $0 \cdot 0 - 1 \cdot 1 = -1$ . The zero block in the corner is what the theorem needs. The correct formulas for general blocks use Schur complements, in Chapter 7.

**Quick check.** Is  $\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ 0 & \mathbf{D} \end{pmatrix} = \det \mathbf{A} \det \mathbf{D}$  meaningful when  $\mathbf{A}$  is  $2 \times 3$  and  $\mathbf{D}$  is  $3 \times 2$  (so the whole matrix is  $5 \times 5$ )?

**Solution.** No. The theorem requires  $\mathbf{A}$  and  $\mathbf{D}$  square; here  $\det \mathbf{A}$  and  $\det \mathbf{D}$  are undefined. In fact the whole determinant is 0 whatever the entries: the first 3 columns have their non-zero entries only in the first 2 rows, so they are linearly dependent in a 2-dimensional space, and a determinant with dependent columns is 0 (Theorem 6.2.1).

## 6.8.4 Rank-one updates

Changing a matrix by a rank-one matrix  $\mathbf{u}\mathbf{v}^\top$  is the smallest possible change in the sense of rank, and it happens often: updating one observation in statistics, modifying one row of a system, or adding the same number to every entry. The determinant of the updated identity turns out to be a single dot product.

### Theorem 6.8.5 — Matrix Determinant Lemma

Let  $\mathbf{u}, \mathbf{v} \in F^n$ . Then

$$\det(\mathbf{I}_n + \mathbf{u}\mathbf{v}^\top) = 1 + \mathbf{v}^\top \mathbf{u}.$$

*Idea.* Border  $\mathbf{I}_n + \mathbf{u}\mathbf{v}^\top$  by an extra column  $\mathbf{u}$  and an extra row  $(\mathbf{0}^\top, 1)$ , giving a block triangular matrix  $\mathbf{M}$  with the same determinant. Then clear the update by elimination with blocks. Rows: add  $\mathbf{v}^\top$  times the top  $n$  rows to the new bottom row (multiply by  $\mathbf{L}$  on the left). Columns: subtract the last column times  $\mathbf{v}^\top$  from the first  $n$  columns (multiply by  $\mathbf{L}'$  on the right). The factors  $\mathbf{L}$ ,  $\mathbf{L}'$  have determinant 1, and the result is block triangular with  $\mathbf{I}_n$  and the  $1 \times 1$  block  $1 + \mathbf{v}^\top \mathbf{u}$  on the diagonal.

*Proof.* Consider the  $(n+1) \times (n+1)$  block matrices

$$\mathbf{L} = \begin{pmatrix} \mathbf{I}_n & \mathbf{0} \\ \mathbf{v}^\top & 1 \end{pmatrix}, \quad \mathbf{M} = \begin{pmatrix} \mathbf{I}_n + \mathbf{u}\mathbf{v}^\top & \mathbf{u} \\ \mathbf{0}^\top & 1 \end{pmatrix}, \quad \mathbf{L}' = \begin{pmatrix} \mathbf{I}_n & \mathbf{0} \\ -\mathbf{v}^\top & 1 \end{pmatrix}.$$

Splitting the sum  $\sum_j$  in Theorem 0.9.2 (1) into the indices  $j \leq n$  and  $j = n+1$  shows that matrices partitioned in this way multiply block by block, like  $2 \times 2$  matrices whose entries are blocks. Hence

$$\mathbf{LM} = \begin{pmatrix} \mathbf{I}_n + \mathbf{u}\mathbf{v}^\top & \mathbf{u} \\ \mathbf{v}^\top + (\mathbf{v}^\top \mathbf{u})\mathbf{v}^\top & \mathbf{v}^\top \mathbf{u} + 1 \end{pmatrix},$$

where we used  $\mathbf{v}^\top(\mathbf{u}\mathbf{v}^\top) = (\mathbf{v}^\top \mathbf{u})\mathbf{v}^\top$  by associativity,  $\mathbf{v}^\top \mathbf{u}$  being a  $1 \times 1$  matrix, that is, a scalar. Multiplying on the right by  $\mathbf{L}'$ ,

$$\mathbf{LML}' = \begin{pmatrix} \mathbf{I}_n + \mathbf{u}\mathbf{v}^\top - \mathbf{u}\mathbf{v}^\top & \mathbf{u} \\ \mathbf{v}^\top + (\mathbf{v}^\top \mathbf{u})\mathbf{v}^\top - (1 + \mathbf{v}^\top \mathbf{u})\mathbf{v}^\top & 1 + \mathbf{v}^\top \mathbf{u} \end{pmatrix} = \begin{pmatrix} \mathbf{I}_n & \mathbf{u} \\ \mathbf{0}^\top & 1 + \mathbf{v}^\top \mathbf{u} \end{pmatrix}.$$

By Theorem 6.8.4,  $\det \mathbf{L} = \det \mathbf{I}_n \cdot 1 = 1$ ,  $\det \mathbf{L}' = 1$ ,  $\det \mathbf{M} = \det(\mathbf{I}_n + \mathbf{u}\mathbf{v}^\top) \cdot 1$ , and  $\det(\mathbf{LML}') = \det \mathbf{I}_n (1 + \mathbf{v}^\top \mathbf{u}) = 1 + \mathbf{v}^\top \mathbf{u}$ . By Theorem 6.5.4,  $\det(\mathbf{LML}') = \det \mathbf{L} \det \mathbf{M} \det \mathbf{L}' = \det(\mathbf{I}_n + \mathbf{u}\mathbf{v}^\top)$ . This proves the lemma. ■

### Example. ( $a$ on the Diagonal, 1 Elsewhere)

Let  $a \in \mathbb{R}$  and  $n \geq 1$ , and let  $\mathbf{A}_n \in M_n(\mathbb{R})$  have every diagonal entry  $a$  and every other entry 1. Find  $\det \mathbf{A}_n$ .

*Solution.* Let  $\mathbf{1} = (1, \dots, 1) \in \mathbb{R}^n$ . Every entry of  $\mathbf{1}\mathbf{1}^\top$  is 1, so  $\mathbf{A}_n = (a-1)\mathbf{I}_n + \mathbf{1}\mathbf{1}^\top$ .

*Case  $a \neq 1$ .* Then  $\mathbf{A}_n = (a-1)(\mathbf{I}_n + \mathbf{u}\mathbf{1}^\top)$  with  $\mathbf{u} = \frac{1}{a-1}\mathbf{1}$ . Scaling an  $n \times n$  matrix by  $c$  multiplies its determinant by  $c^n$  (multilinearity in each of the  $n$  columns), so by Theorem 6.8.5

$$\det \mathbf{A}_n = (a-1)^n \left(1 + \frac{n}{a-1}\right) = (a-1)^{n-1}(a-1+n),$$

since  $\mathbf{1}^\top \mathbf{u} = \frac{n}{a-1}$ .

*Case  $a = 1$ .* For  $n \geq 2$ , all columns are equal, so  $\det \mathbf{A}_n = 0$ , which the formula also gives. For  $n = 1$ ,  $\det \mathbf{A}_1 = 1$ , and the formula gives  $(a-1)^0 \cdot 1 = 1$ .

So  $\det \mathbf{A}_n = (a-1)^{n-1}(a+n-1)$  for every  $a$ . For  $n = 3$  this is  $(a-1)^2(a+2)$ ; with  $a = 3$  it gives  $4 \cdot 5 = 20$ .

*Warning.* The lemma is about  $\mathbf{I}_n + \mathbf{u}\mathbf{v}^\top$ , not  $\mathbf{u}\mathbf{v}^\top$  or  $\mathbf{u}^\top\mathbf{v}$ . The rank-one matrix  $\mathbf{u}\mathbf{v}^\top$  is  $n \times n$  and, for  $n \geq 2$ , has determinant 0; the scalar  $\mathbf{v}^\top\mathbf{u}$  is its trace. Do not confuse  $\det(\mathbf{I}_n + \mathbf{u}\mathbf{v}^\top) = 1 + \mathbf{v}^\top\mathbf{u}$  with  $\det(\mathbf{I}_n) + \det(\mathbf{u}\mathbf{v}^\top) = 1$  (for  $n \geq 2$ ), which is false whenever  $\mathbf{v}^\top\mathbf{u} \neq 0$ : for  $\mathbf{u} = \mathbf{v} = \mathbf{e}_1$ ,  $\mathbf{I}_n + \mathbf{e}_1\mathbf{e}_1^\top = \text{diag}(2, 1, \dots, 1)$  has determinant 2.

## 6.8.5 Exercises

### 6.8.5.1 A. Check your understanding

#### Exercise A1.

- State the Vandermonde determinant formula, including the order of the factors.
- True or false:  $\mathbf{V}(x_1, \dots, x_n)$  is invertible whenever  $x_1, \dots, x_n$  are non-zero. Justify your answer.
- Write down the recurrence for the determinant of a tridiagonal matrix, and say which row the proof expands along.
- True or false:  $\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} = \det \mathbf{A} \det \mathbf{D}$  whenever  $\mathbf{A}$  and  $\mathbf{D}$  are square and  $\mathbf{B} = \mathbf{0}$ . Justify your answer.
- True or false:  $\det(\mathbf{I}_n + \mathbf{u}\mathbf{v}^\top) = \det(\mathbf{I}_n + \mathbf{v}\mathbf{u}^\top)$  for all  $\mathbf{u}, \mathbf{v} \in F^n$ . Justify your answer.

*Solution.*

- $\det \mathbf{V}(x_1, \dots, x_n) = \prod_{1 \leq i < j \leq n} (x_j - x_i)$ , where  $\mathbf{V}$  has  $(i, j)$ -entry  $x_i^{j-1}$  ([Theorem 6.8.1](#)).
- False.  $\mathbf{V}(1, 1) = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$  has non-zero nodes and determinant 0. The condition is that the nodes are **distinct** ([Corollary 6.8.2](#)); 0 is allowed as a node.
- $D_k = a_k D_{k-1} - b_{k-1} c_{k-1} D_{k-2}$ , with  $D_0 = 1$ ,  $D_1 = a_1$  ([Theorem 6.8.3](#)). The proof expands along the last row, which has only two non-zero entries.
- True. With  $\mathbf{B} = \mathbf{0}$  the matrix is lower block triangular, and [Theorem 6.8.4](#) applies.
- True. Both equal  $1 + \mathbf{v}^\top\mathbf{u} = 1 + \mathbf{u}^\top\mathbf{v}$  by [Theorem 6.8.5](#), since  $\mathbf{v}^\top\mathbf{u} = \sum_i v_i u_i = \mathbf{u}^\top\mathbf{v}$ .

### 6.8.5.2 B. Practice

#### Exercise B1: A $4 \times 4$ Vandermonde determinant.

Compute  $\det \begin{pmatrix} 1 & -1 & 1 & -1 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 \end{pmatrix}$ . Hence decide whether there is exactly one polynomial  $p \in \mathbb{R}[x]_{\leq 3}$  with  $p(-1) = 3$ ,  $p(0) = 1$ ,  $p(1) = 0$ ,  $p(2) = 5$ .

*Solution.* Row  $i$  is  $(1, x_i, x_i^2, x_i^3)$  with nodes  $-1, 0, 1, 2$ , so this is  $\mathbf{V}(-1, 0, 1, 2)$ . By

**Theorem 6.8.1** its determinant is

$$(0 - (-1))(1 - (-1))(2 - (-1))(1 - 0)(2 - 0)(2 - 1) = 1 \cdot 2 \cdot 3 \cdot 1 \cdot 2 \cdot 1 = 12.$$

(Check by expanding along row 2, which is  $(1, 0, 0, 0)$ : the determinant is  $(-1)^{2+1} \det \begin{pmatrix} -1 & 1 & -1 \\ 1 & 1 & 1 \\ 2 & 4 & 8 \end{pmatrix} = -(-1(8 - 4) - 1(8 - 2) - 1(4 - 2)) = -(-4 - 6 - 2) = 12.$ ) Hence the coefficient matrix of the interpolation system is invertible, and there is exactly one  $p \in \mathbb{R}[x]_{\leq 3}$  with the given values.

### Exercise B2: The second-difference matrix.

Let  $\mathbf{K}_n \in M_n(\mathbb{R})$  have 2 on the diagonal,  $-1$  directly above and below it, and 0 elsewhere. Prove that  $\det \mathbf{K}_n = n + 1$ . Hence show that  $\mathbf{K}_n$  is invertible for every  $n$ .

*Solution.* Here  $a_k = 2$  and  $b_{k-1}c_{k-1} = (-1)(-1) = 1$ . By [Theorem 6.8.3](#),  $D_k = 2D_{k-1} - D_{k-2}$  for  $k \geq 2$ , with  $D_0 = 1$ ,  $D_1 = 2$ . We show  $D_k = k + 1$  by strong induction on  $k$  (Theorem 0.2.10). It holds for  $k = 0, 1$ . If it holds for  $k - 1$  and  $k - 2$ , then  $D_k = 2k - (k - 1) = k + 1$ . Hence  $\det \mathbf{K}_n = D_n = n + 1 \neq 0$ , and  $\mathbf{K}_n$  is invertible by Theorem 6.5.2.

### Exercise B3: A rank-one update.

Let  $\mathbf{u} = (1, 2, 3)$  and  $\mathbf{v} = (1, -1, 2)$  in  $\mathbb{Q}^3$ .

- Use [Theorem 6.8.5](#) to find  $\det(\mathbf{I}_3 + \mathbf{u}\mathbf{v}^\top)$ .
- Write out  $\mathbf{I}_3 + \mathbf{u}\mathbf{v}^\top$  and check your answer by elimination.
- Find all  $t \in \mathbb{Q}$  for which  $\mathbf{I}_3 + t\mathbf{u}\mathbf{v}^\top$  is **not** invertible.

*Solution.*

- $\mathbf{v}^\top \mathbf{u} = 1 - 2 + 6 = 5$ , so  $\det(\mathbf{I}_3 + \mathbf{u}\mathbf{v}^\top) = 6$ .

- $\mathbf{u}\mathbf{v}^\top = \begin{pmatrix} 1 & -1 & 2 \\ 2 & -2 & 4 \\ 3 & -3 & 6 \end{pmatrix}$ , so  $\mathbf{I}_3 + \mathbf{u}\mathbf{v}^\top = \begin{pmatrix} 2 & -1 & 2 \\ 2 & -1 & 4 \\ 3 & -3 & 7 \end{pmatrix}$ . Subtract row 1 from row 2, which does not change the determinant (Theorem 6.5.1), to get row 2 equal to  $(0, 0, 2)$ . Expanding along that row,

$$\det = 2 \cdot (-1)^{2+3} \det \begin{pmatrix} 2 & -1 \\ 3 & -3 \end{pmatrix} = -2(-6 + 3) = 6.$$

- $\mathbf{I}_3 + t\mathbf{u}\mathbf{v}^\top = \mathbf{I}_3 + (t\mathbf{u})\mathbf{v}^\top$ , whose determinant is  $1 + t\mathbf{v}^\top \mathbf{u} = 1 + 5t$ . It is zero exactly for  $t = -\frac{1}{5}$ , and by Theorem 6.5.2 that is the only  $t$  for which the matrix is not invertible.

#### 6.8.5.3 C. Going deeper

**Exercise C1: The Cauchy determinant.**

Let  $x_1, \dots, x_n, y_1, \dots, y_n \in F$  with  $x_i + y_j \neq 0$  for all  $i, j$ , and let  $\mathbf{Q}_n \in M_n(F)$  have  $(i, j)$ -entry  $\frac{1}{x_i + y_j}$ . The goal is

$$\det \mathbf{Q}_n = \frac{\prod_{1 \leq i < j \leq n} (x_j - x_i)(y_j - y_i)}{\prod_{i=1}^n \prod_{j=1}^n (x_i + y_j)}.$$

- (a) For  $i < n$ , subtract row  $n$  from row  $i$ . Show that the new  $(i, j)$ -entry is  $\frac{x_n - x_i}{(x_i + y_j)(x_n + y_j)}$ , and deduce that  $\det \mathbf{Q}_n = \frac{\prod_{i < n} (x_n - x_i)}{\prod_j (x_n + y_j)} \det \mathbf{Q}'$ , where  $\mathbf{Q}'$  agrees with  $\mathbf{Q}_n$  in rows  $1, \dots, n-1$  and has last row  $(1, \dots, 1)$ .
- (b) In  $\mathbf{Q}'$ , for  $j < n$ , subtract column  $n$  from column  $j$ . Deduce that  $\det \mathbf{Q}' = \frac{\prod_{j < n} (y_n - y_j)}{\prod_{i < n} (x_i + y_n)} \det \mathbf{Q}_{n-1}$ , where  $\mathbf{Q}_{n-1}$  is built from  $x_1, \dots, x_{n-1}$  and  $y_1, \dots, y_{n-1}$ .
- (c) Prove the formula by induction on  $n$ .
- (d) The **Hilbert matrix**  $\mathbf{H}_n$  has  $(i, j)$ -entry  $\frac{1}{i+j-1}$ . Show that it is a Cauchy matrix, and use (c) to compute  $\det \mathbf{H}_3$ .

*Hint: in (a) and (b), factor scalars out of whole rows and columns using multilinearity.*

*Solution.*

- (a) For  $i < n$ ,  $\frac{1}{x_i + y_j} - \frac{1}{x_n + y_j} = \frac{(x_n + y_j) - (x_i + y_j)}{(x_i + y_j)(x_n + y_j)} = \frac{x_n - x_i}{(x_i + y_j)(x_n + y_j)}$ . These row operations do not change the determinant (Theorem 6.5.1). In the new matrix, row  $i < n$  has the factor  $x_n - x_i$  in every entry, and column  $j$  has the factor  $\frac{1}{x_n + y_j}$  in every entry (in row  $n$ ,  $\frac{1}{x_n + y_j} = \frac{1}{x_n + y_j} \cdot 1$ ). Pulling these scalars out of each row and column by multilinearity (Theorem 6.5.1, in its row and column forms) leaves the matrix  $\mathbf{Q}'$  with entries  $\frac{1}{x_i + y_j}$  in rows  $i < n$  and 1 in row  $n$ . This gives the stated formula.
- (b) For  $j < n$ , subtracting column  $n$  from column  $j$  leaves the determinant unchanged, makes the last row  $(0, \dots, 0, 1)$ , and for  $i < n$  gives  $\frac{1}{x_i + y_j} - \frac{1}{x_i + y_n} = \frac{y_n - y_j}{(x_i + y_j)(x_i + y_n)}$ . Pull  $y_n - y_j$  out of column  $j$  ( $j < n$ ) and  $\frac{1}{x_i + y_n}$  out of row  $i$  ( $i < n$ ); the last row  $(0, \dots, 0, 1)$  is unaffected by the column factors because its first  $n-1$  entries are 0, and row factors do not touch row  $n$ . What remains has  $\frac{1}{x_i + y_j}$  in positions  $i, j < n$ , entries  $(x_i + y_n) \cdot \frac{1}{x_i + y_n} = 1$  in column  $n$  above the corner, and last row  $(0, \dots, 0, 1)$ . Expanding along the last row (Theorem 6.6.1) gives  $\det \mathbf{Q}_{n-1}$ , with sign  $(-1)^{2n} = 1$ .
- (c) For  $n = 1$ ,  $\det \mathbf{Q}_1 = \frac{1}{x_1 + y_1}$ , which is the formula with empty products upstairs. For  $n \geq 2$ , combining (a) and (b),

$$\det \mathbf{Q}_n = \frac{\prod_{i < n} (x_n - x_i)(y_n - y_i)}{\prod_{j \leq n} (x_n + y_j) \prod_{i < n} (x_i + y_n)} \det \mathbf{Q}_{n-1}.$$

The numerator supplies exactly the factors of  $\prod_{i < j} (x_j - x_i)(y_j - y_i)$  with  $j = n$ , and the denominator supplies exactly the factors  $x_i + y_j$  with  $\max(i, j) = n$ , each once. By the induction hypothesis,  $\det \mathbf{Q}_{n-1}$  supplies all the remaining factors. This proves the formula.

- (d) With  $x_i = i$  and  $y_j = j - 1$ ,  $x_i + y_j = i + j - 1 \geq 1$ , so  $\mathbf{H}_n$  is a Cauchy matrix over  $\mathbb{Q}$ . For  $n = 3$ , the numerator is  $((2-1)(3-1)(3-2))^2 = 4$ , since  $y_j - y_i = j - i = x_j - x_i$ . The denominator is the product of all  $i + j - 1$ : the rows give  $1 \cdot 2 \cdot 3 = 6$ ,  $2 \cdot 3 \cdot 4 = 24$ ,  $3 \cdot 4 \cdot 5 = 60$ , in total 8640. Hence  $\det \mathbf{H}_3 = \frac{4}{8640} = \frac{1}{2160}$ . (In general  $\det \mathbf{H}_n = c_n^4 / c_{2n}$  with  $c_n = \prod_{k=1}^{n-1} k!$ ; we state this without proof. These determinants shrink extremely fast, which is why Hilbert matrices are a standard test of numerical methods.)

### Exercise C2: A circulant determinant over $\mathbb{C}$ .

Let  $a, b, c \in \mathbb{C}$ , let  $\omega = e^{2\pi i/3}$ , and let

$$\mathbf{Z} = \begin{pmatrix} a & b & c \\ c & a & b \\ b & c & a \end{pmatrix}.$$

- (a) For  $k = 0, 1, 2$ , let  $\mathbf{w}_k = (1, \omega^k, \omega^{2k})$  and  $\lambda_k = a + b\omega^k + c\omega^{2k}$ . Show that  $\mathbf{Z}\mathbf{w}_k = \lambda_k \mathbf{w}_k$ .
- (b) Let  $\mathbf{W}$  be the matrix with columns  $\mathbf{w}_0, \mathbf{w}_1, \mathbf{w}_2$ . Show that  $\mathbf{Z}\mathbf{W} = \mathbf{W} \operatorname{diag}(\lambda_0, \lambda_1, \lambda_2)$  and that  $\det \mathbf{W} \neq 0$ . Deduce  $\det \mathbf{Z} = \lambda_0 \lambda_1 \lambda_2$ .
- (c) Using  $\omega^3 = 1$  and  $1 + \omega + \omega^2 = 0$ , deduce  $\det \mathbf{Z} = a^3 + b^3 + c^3 - 3abc$ , and check this against the Leibniz formula.

Circulant matrices of every size are treated in Chapter 11.

*Solution.*

- (a) By Example (Roots of Unity),  $\omega^3 = 1$ , so  $\omega^{-k} = \omega^{2k}$  and  $\omega^{-2k} = \omega^k$ . Row 1 of  $\mathbf{Z}\mathbf{w}_k$  is  $a + b\omega^k + c\omega^{2k} = \lambda_k$ . Row 2 is  $c + a\omega^k + b\omega^{2k} = \omega^k(c\omega^{-k} + a + b\omega^k) = \omega^k(a + b\omega^k + c\omega^{2k}) = \omega^k \lambda_k$ . Row 3 is  $b + c\omega^k + a\omega^{2k} = \omega^{2k}(b\omega^{-2k} + c\omega^{-k} + a) = \omega^{2k} \lambda_k$ . Hence  $\mathbf{Z}\mathbf{w}_k = \lambda_k \mathbf{w}_k$ .
- (b) By Theorem 0.9.2, column  $k$  of  $\mathbf{Z}\mathbf{W}$  is  $\mathbf{Z}\mathbf{w}_k = \lambda_k \mathbf{w}_k$ , which is column  $k$  of  $\mathbf{W} \operatorname{diag}(\lambda_0, \lambda_1, \lambda_2)$ . The  $(i, k)$ -entry of  $\mathbf{W}$  is  $(\omega^k)^{i-1}$ , so  $\mathbf{W} = \mathbf{V}(1, \omega, \omega^2)^\top$ . The nodes  $1, \omega, \omega^2$  are distinct (they are  $e^{2\pi i k/3}$  for  $k = 0, 1, 2$ ), so  $\det \mathbf{W} = \det \mathbf{V}(1, \omega, \omega^2)^\top \neq 0$  by Theorem 6.4.5 and Corollary 6.8.2. By Theorem 6.5.4 and Theorem 6.4.6,  $\det \mathbf{Z} \det \mathbf{W} = \det \mathbf{W} \lambda_0 \lambda_1 \lambda_2$ , and canceling  $\det \mathbf{W} \neq 0$  gives  $\det \mathbf{Z} = \lambda_0 \lambda_1 \lambda_2$ .
- (c) Multiply out  $\lambda_1 \lambda_2 = (a + b\omega + c\omega^2)(a + b\omega^2 + c\omega)$ . The nine terms are  $a^2, ab\omega^2 + ab\omega, ac\omega + ac\omega^2, b^2\omega^3, bc\omega^2, bc\omega^4 = bc\omega, c^2\omega^3$ . Using  $\omega^3 = 1$  and  $\omega + \omega^2 = -1$ ,

$$\lambda_1 \lambda_2 = a^2 + b^2 + c^2 - ab - ac - bc.$$

Then  $\det \mathbf{Z} = (a + b + c)(a^2 + b^2 + c^2 - ab - ac - bc) = a^3 + b^3 + c^3 - 3abc$ , expanding and canceling. The Leibniz formula for  $\mathbf{Z}$  gives the three even terms  $a \cdot a \cdot a, c \cdot c \cdot c$  (from  $\sigma = (1 \ 2 \ 3)$ ) and  $b \cdot b \cdot b$ , and the three odd terms, each equal to  $abc$  with sign  $-$ : the same answer.

### Exercise C3: Rank-one updates of an invertible matrix.

Let  $\mathbf{A} \in M_n(F)$  be invertible and  $\mathbf{u}, \mathbf{v} \in F^n$ .

- (a) Prove that  $\det(\mathbf{A} + \mathbf{u}\mathbf{v}^\top) = \det \mathbf{A} (1 + \mathbf{v}^\top \mathbf{A}^{-1} \mathbf{u})$ .
- (b) Deduce that  $\mathbf{A} + \mathbf{u}\mathbf{v}^\top$  is invertible if and only if  $\mathbf{v}^\top \mathbf{A}^{-1} \mathbf{u} \neq -1$ .
- (c) Give an example over  $\mathbb{Q}$  with  $n = 2$  where  $\mathbf{A}$  is invertible,  $\mathbf{u}\mathbf{v}^\top \neq 0$ , and  $\mathbf{A} + \mathbf{u}\mathbf{v}^\top$  is **not** invertible.

*Solution.*

- (a) Since  $\mathbf{A}$  is invertible,  $\mathbf{A} + \mathbf{u}\mathbf{v}^\top = \mathbf{A}(\mathbf{I}_n + \mathbf{A}^{-1}\mathbf{u}\mathbf{v}^\top) = \mathbf{A}(\mathbf{I}_n + (\mathbf{A}^{-1}\mathbf{u})\mathbf{v}^\top)$  by Theorem 0.9.4. By Theorem 6.5.4 and [Theorem 6.8.5](#) applied to the vectors  $\mathbf{A}^{-1}\mathbf{u}$  and  $\mathbf{v}$ ,  $\det(\mathbf{A} + \mathbf{u}\mathbf{v}^\top) = \det \mathbf{A} (1 + \mathbf{v}^\top \mathbf{A}^{-1} \mathbf{u})$ .
- (b)  $\det \mathbf{A} \neq 0$  by Theorem 6.5.2, so  $\det(\mathbf{A} + \mathbf{u}\mathbf{v}^\top) \neq 0$  exactly when  $1 + \mathbf{v}^\top \mathbf{A}^{-1} \mathbf{u} \neq 0$ . Apply Theorem 6.5.2 again.
- (c) Take  $\mathbf{A} = \mathbf{I}_2$ ,  $\mathbf{u} = (-1, 0)$ ,  $\mathbf{v} = (1, 0)$ . Then  $\mathbf{v}^\top \mathbf{u} = -1$ , and indeed  $\mathbf{A} + \mathbf{u}\mathbf{v}^\top = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$  is not invertible.